

TECHNIQUE

AFTER READING THE *Climbing Bible* you'll be familiar with the fundamentals which underlie all climbing movement. You should also know which specific techniques you can use to move around on the wall in a controlled and efficient manner. In this chapter we cover different exercises to train these fundamentals and specific techniques. We also show you efficient exercises for training footwork and grip positions. We begin with the easiest exercises and move on to gradually more challenging ones, so start working on your technique with the first one and work your way up from there.

Before you begin, we want to repeat some of the most important things you should keep in mind when practising technique. First and foremost, it is important that you reserve some of your training sessions – where you are rested and motivated to learn – for focusing only on technique. These exercises require you to concentrate, so avoid practising technique when you are tired. You can practise technique in the same sessions as when you train strength or endurance, but it should always come before the hard physical training.

You should also have separate focus areas for each session. For optimal progression it is important to target what you feel are your weaknesses as a climber. However, you should further develop what you are already good at as well, so continue to cultivate your strengths while working on your weaknesses. If you're unsure of your strengths and weaknesses, you can ask a climbing partner or get help from an experienced coach.

When the session starts, it's important that you have a sense of how the exercise should be carried out and how you're performing it. By first observing others and then filming yourself while doing the exercise, it will be easier for you to get a sense for how you're doing and get direct feedback on your performance. If you also have someone you can discuss the exercise with, the chances of learning will be greater. A climbing partner with more experience than yourself or an experienced climbing coach are the best people with which to discuss the exercise.

INTENSITY

In most sports, athletes train technique at low intensity levels. In climbing, however, we have a tendency to always try our hardest to top a boulder problem or route in every single session. This is not optimal! By conducting specific sessions at lower intensities and focusing on proper execution, you'll soon become much better at moving around on the wall.



PHOTO: STIAN CHRISTOPHERSEN

Joakim Louis Sæther on *Maraton*, Norway's first F8b. *Maraton* is located at the Demperhylla sector at Damtjern. The crag is known for vertical walls, small holds and technically demanding climbing.

FOOTWORK

YOUR LEGS AND feet can and should take a large part of the load that is necessary to keep you on the wall. To be able to do this, you need to be precise when placing your feet on the holds to ensure that they don't slip off, and you should lower your body such that as much of your weight as possible is distributed on to your feet and off of your upper body. The following pages feature some exercises to target exactly this.



PHOTO: PETTER SLUNGAARD KRISTENSEN

Jarl Gåsvær sending *Storpillaren* (N7, approx. E5 6a) on Vågakallen, Lofoten, Norway.

CHOOSING SHOES

As a general rule, you shouldn't wear worn-out shoes when working on refining your footwork. Trusting your feet is something that will come naturally when you're actually standing on holds that you consider difficult to stand on; the likelihood of this happening with worn-out shoes isn't great. On the following pages we highlight the advantages of soft or stiff shoes where choosing one or the other has an impact on the exercise, but for most of the exercises any good all-round shoe will do.



LOOK UP – LOOK DOWN

Many climbers, especially beginners, forget to look at where they are placing their feet. Therefore, make it a rule to always look down at your foot when you're placing it on a hold. It's important that you do not look up before you have actually placed your foot exactly where you want it. When your foot is in place you can look up and move to the next handhold. Your gaze should therefore alternate up and down every time you move a foot or a hand. Try climbing a route where you consistently move your gaze to the foot or hand about to be moved. If you forget, reverse the move and do it again.



This can quickly become rather pumpy, so begin with routes well within your ability level.

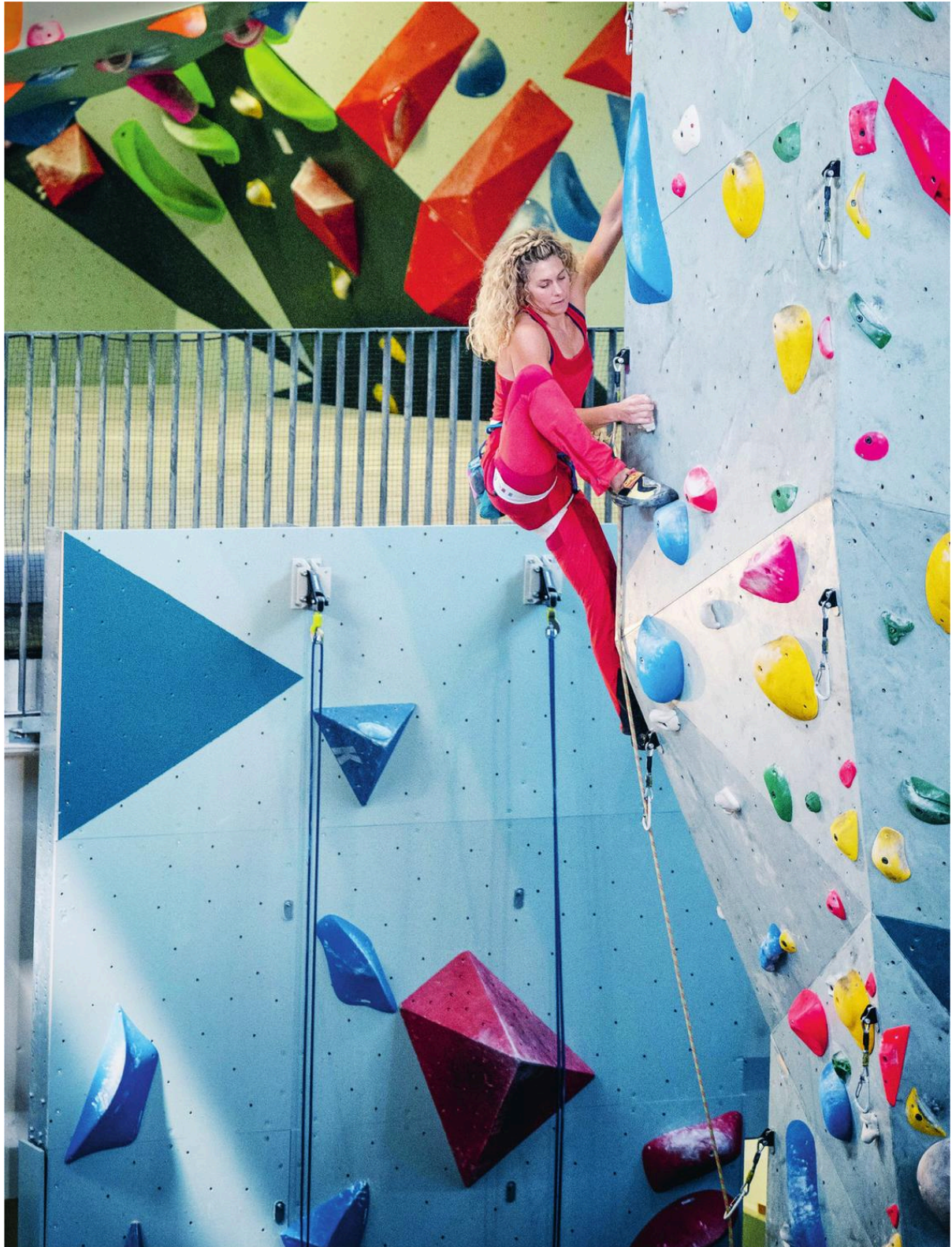


BABY STEPS

To help practise your footwork, try making three moves with your feet for every individual hand move. This is perhaps unnecessary to climb a route or a boulder problem, but it's a good way to practise precision footwork and it makes you more aware of how important it is to actually look at your feet when you are placing them on a foothold. It's a good idea to start with routes with large footholds before progressing to smaller and smaller holds.



It can be a good idea to choose lines on the wall with multiple routes as this way you can choose to use footholds from another route if necessary.



HIGH STEPS

As a climber, you're bound to come across high steps, so you might as well start practising them from the get-go. To practise efficiently, choose a route you can do quite easily and place your feet as high as you can for every move. High steps require flexibility, but it's possible to cheat your way up by moving your hips out and away from the wall, as long as the handholds are good enough. However, climbing on slopers inhibits you from moving your hips too far out from the wall and therefore forces you to work on hip mobility when placing your feet high.



Stiffer shoes are better when doing high steps to small footholds.



DOWNCLIMBING

Downclimbing forces you to look at your feet when you are placing them, so try climbing up a route and then back down again. Pair up with a climbing partner and alternate who climbs and who belays. You can also do this session on an auto belay, but this isn't ideal, as the auto belay is constantly pulling upwards while you're downclimbing.



Start with easy, vertical routes which are well within your limit as downclimbing is more pumpy than you might think.



NINJA FEET

Try climbing a route or boulder problem that's well within your limit and without making a sound when placing your feet on the footholds. To be able to do this, you will need to look at your feet and be very precise when placing them on the holds. This will also train your balance, as you will need to be in a position that gives you enough time to place your feet properly and with precision.

If you find placing your feet precisely and silently difficult, instead try practising while still on the ground. Choose different footholds near the ground – preferably holds with different shapes – and practise placing one foot while keeping the other on the ground. Keep your balance by holding on to holds on the wall with your hands as this mimics the situation you'll face while actually climbing.

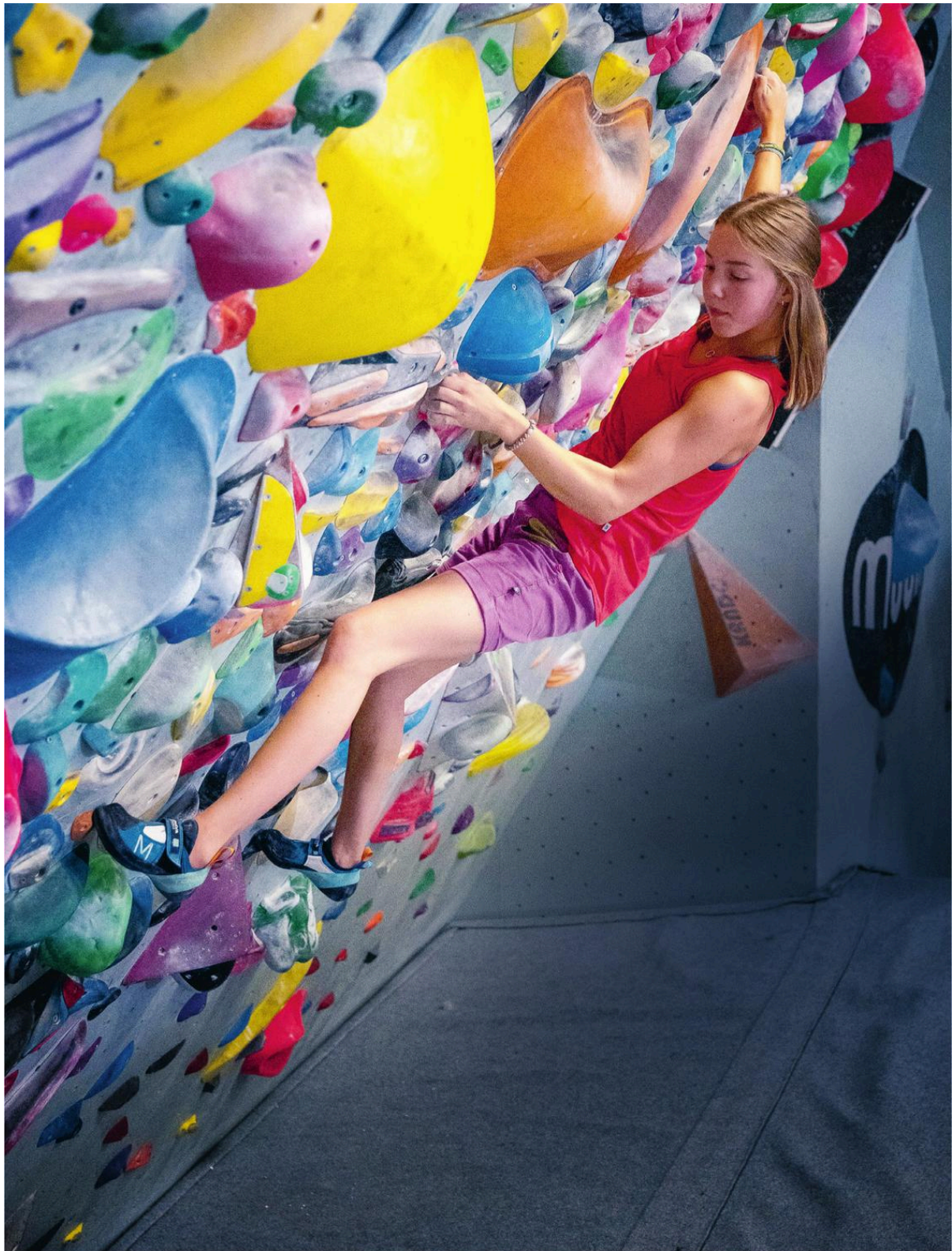


Use softer shoes to begin with as it's easier to precisely place a softer shoe. They also make less sound, as the softer rubber more easily conforms to the shape of the holds.



STICKY FEET

On many footholds there might be only the tiniest of grooves in which to place your foot. When placing your foot on these holds there's little room to adjust your foot after placing it. This means you have to be very precise when placing the tip of your shoe on the hold. To practise this, you can climb routes where you don't allow yourself to move or twist your foot to either side once it is placed on a hold.



CHIPS ONLY

The style of route setting in climbing walls differs from one to the other, but if you have the option to climb routes with small screw-on footholds or chips only, this is a really good way to train precision footwork.* Standing on small footholds demands even more precision to hit that tiny dimple or protrusion.

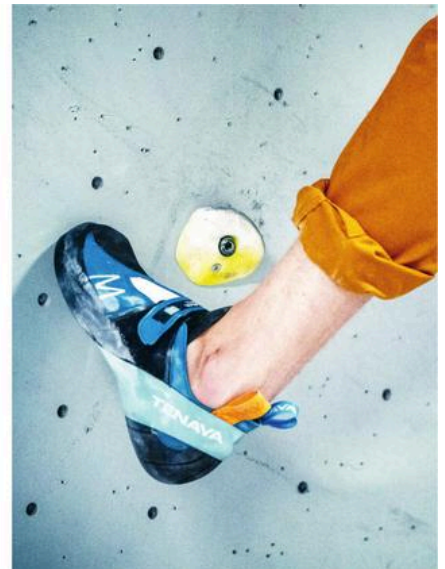
* Chips are those small holds set with regular wood screws instead of bolts like normal holds, and intended to be used exclusively as footholds. Also known as foot dinks, jibs or just screw-ons.



Use your creativity and climb routes where you only use the small footholds (if this is an option). You could also choose to stand only on the tiny dimple on the side of a larger hold, or all the way to the left or right on a larger edge, for example.



How to place your foot correctly.



Your foot can twist off the hold when placed incorrectly.

USE THE TIP

Rookie climbers will tend to place the middle of their foot on a foothold. This is unfortunate as a climbing shoe loses many of its properties when placed in this way. In addition to your climbing being less precise, you will lose the ability to twist your body around the foothold without your foot slipping off the hold. In most cases you should use the tip of your shoe to stand on footholds. You will sometimes use the sides of the shoe, but even then it's still the inside or outside edge of the tip of the shoe which you're using.

A good exercise for practising this is to traverse a wall with lots of holds and observe what happens when you place the inside or outside of your foot on the hold, and not the tip. In many cases you'll fall when you move on as your foot will twist off the hold.



INTERMEDIATE STEPS

In order to stand securely on slopey holds, volumes and walls it's important to be able to smear your feet on the holds. And to smear on the footholds you need to lower your heel to maximise the contact surface between the shoe and the hold/wall. A good way to practise this is to stand directly on to the wall (i.e. not on a foothold) with one foot before placing your other foot on the next foothold. Choose routes or boulder problems well within your limit and only use walls with friction. If the walls in your gym are smoothly painted, it'll be near impossible to complete this exercise. If this is the case, you can try to do the same exercise every time there's a volume on the route or boulder problem. There's usually always friction on these.



To understand the importance of lowering your heel, try playing around with a couple of good handholds and a wall with friction: start with your arms straight and your heel lowered. Then pull up with your arms and notice how as your heel lifts the chance of your foot slipping increases.



MATCHING FEET

When climbing, you'll often need to match your feet, which means switching your feet on the same foothold. This is an important technique, particularly when traversing. There are several ways to match your feet. If the hold is relatively large, you can place one foot to the side so that there's enough room for your other foot as well. This doesn't require more than a bit of planning and simple precision. If there's not enough room to do it this way you're left with two choices.

You can slide one foot to the side and then slide the other after it, such that they both fit on to the hold. If the foothold is really small and there's not enough room for your other foot until the first one is off, the sliding technique could still work if you're precise enough that your foot slides on to the hold just as the first one slides off.

The alternative is somewhat more risky, but with good coordination it's very effective. It involves doing a small 'jump' with your foot, switching feet while in the air and then landing on the other foot. 'Jump' might be too strong of a word as you're not trying to jump high into the air, but just high enough for you to sneak one foot underneath the other. We're only talking a few centimetres here. You'll also be able to slow the speed of your jump if the handholds allow it.

We suggest you practise matching feet down low on the bouldering wall, as here you'll usually find lots of good holds. Try traversing the whole wall and match your feet on all the footholds you use. Vary the wall angle and size of the footholds. Use poorer and poorer handholds, so that you can't just 'cheat' by pulling up with your arms.



Train all forms of foot matching – it will definitely come in handy!



STRONG FEET

Keeping your feet on the wall when it's steep is very challenging. It requires a combination of strength and technique. The best exercise for targeting this skill is to climb steep walls while focusing on keeping your feet on the wall as you're moving up. You should stand on your toes and avoid using heel and toe hooks. Maintain tension throughout your body – from your hands to your feet. Chances are you'll fall the second you release the tension. It's important to start with good footholds and shorter moves and then progressively up the difficulty by increasing the distance between the holds and by using smaller footholds. The exercise becomes significantly harder when the movements speed up, so start by doing static moves only, and then gradually make your moves more dynamic.

Use aggressive shoes with a downturned toe box. They will allow you to maintain pressure on the footholds when the wall is steep. Think of your toes as hooks that grip behind the edges of the holds. On



small footholds, there might not be any edge to catch, but we've found that it helps to keep the same focus.



TRICKSTER

The opposite of the previous exercise, now it's time to climb steep routes or boulder problems using all the tricks of the trade so as not to cut loose with your feet. Toe and heel hooks make it significantly easier to keep your feet on the wall when things get steep, as in many cases they remove the need for maintaining tension throughout your body.

Start with shorter moves and climb statically. Gradually increase the lengths of the moves, making them more dynamic. Walls ranging from 40 degrees overhanging to full-on roofs are great for this exercise.



You can 'bicycle' (i.e. stand on top of a hold while toe hooking on the same hold) on almost any hold when it's steep. Even tiny foot chips can be used for toe hooking to take some of the load off in overhangs.

GRIP POSITIONS

IT'S IMPORTANT TO be able to use different grip positions when we're climbing. In *The Climbing Bible* we defined open hand, pinch, half crimp and crimp. We pointed out that the more aggressive grip positions, like the

half crimp and crimp, allow for larger and more explosive movements, while the open-handed grip lends itself to a more precise and careful style of climbing. In addition, a hold will often dictate the best way of holding it. To be versatile climbers we must train the different grip positions, and the following pages demonstrate some exercises for doing just that.



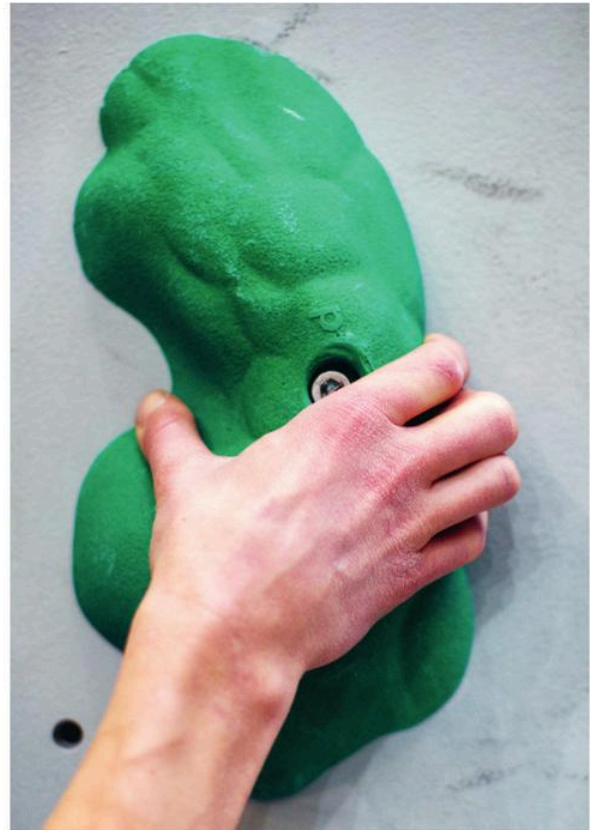
PHOTO: THILO SCHRÖTER

Martin Moberg pinching hard with his left hand in order to send *Eye of Sauron* (Font 7c+), Rocklands, South Africa.

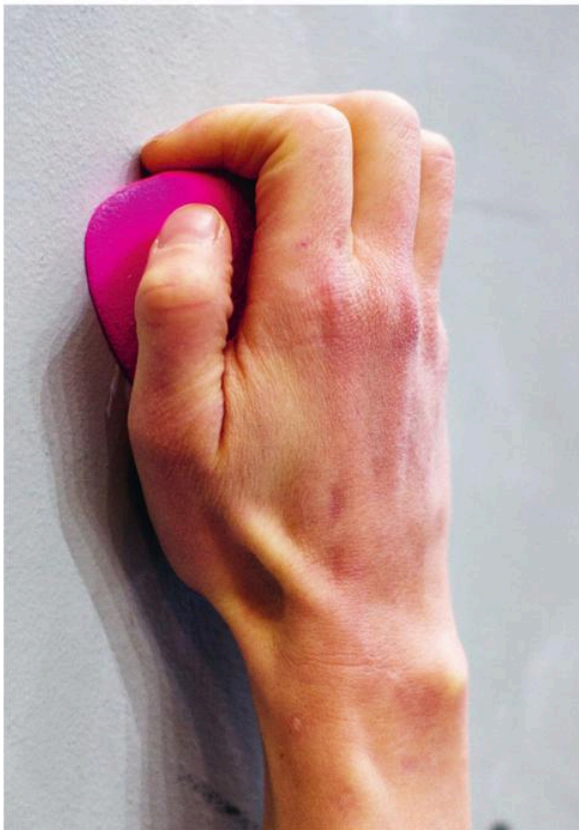
GRIP POSITIONS



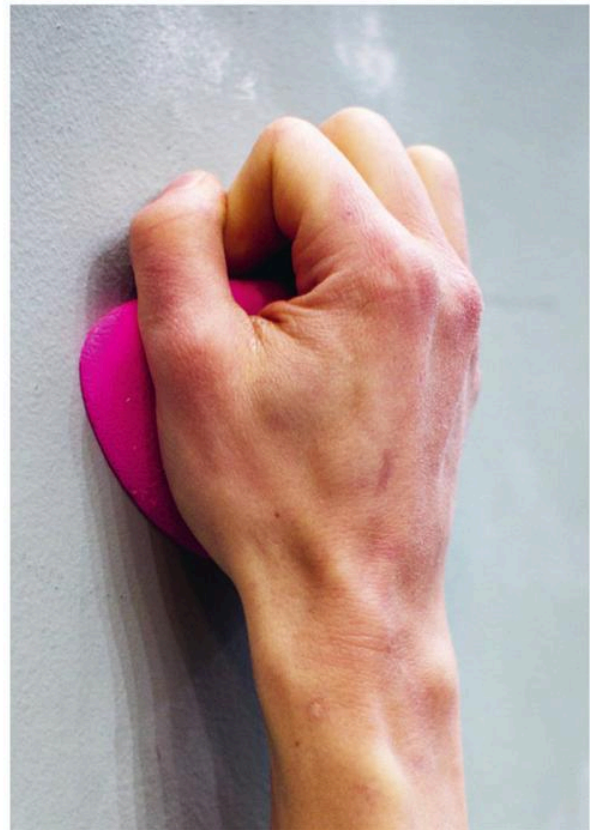
Open hand



Pinch



Half crimp



Crimp



LET THE HOLD DECIDE

Most of us have a go-to grip position we prefer and resort to when we get pumped; by doing this we'll get stronger at using our preferred grip position, but we also need to get stronger in the other positions as well. To practise this, climb a route or boulder problem you've sent before. Use an open-handed grip on all the pockets and good holds, use a half crimp on all edges, and pinch every pinchable hold. Be strict, and ask a climbing buddy to be a judge and let you know if you're cheating.



PRACTISING YOUR CRIMPS

In our experience, many climbers struggle with the half crimp and crimp when starting their climbing careers. The most common grip position for beginners is the open-handed grip. This is a good thing, as this grip minimises the risk of injuries, but it also limits your climbing. By simply standing on the ground while training the half crimp and crimp on an edge, you have a safe and easy method of getting comfortable with the grip positions before putting them to good use on the wall. When you have placed the grip properly, you can bend your knees slightly and start to load the grip – just enough to give you an idea of how it will feel to use the grip while climbing. Hanging from the edge can be trained specifically through deadhangs and is described further in [chapter 2](#).

Be careful with the crimp. This grip is an important weapon when needed, but it should not be used too much during training.



BALANCE

GOOD BALANCE ON the wall is perhaps the most important skill to master in order to conserve energy while climbing. Just think how much extra power you need to cling on to the wall when you lose balance, or how taxing it is to climb routes where you're constantly pushed off balance – like when you're laybacking. In *The Climbing Bible* we defined being in balance as having your centre of gravity centred on your area of support. In simple terms this means having your hips balanced between your feet. Wide foot placements make it easier to maintain balance. It's the same principle as when standing on the ground; if you're standing with wide feet you won't tip over if your buddy gives you a nudge, but with your feet together it's easy to lose balance and fall. Here are some good exercises to train your balance on the wall.



Balance is best trained on slabs and vertical or slightly overhanging walls. Use poor handholds, preferably slopers.



PHOTO: BJØRNAR SMESTAD

Stian Christophersen sending *Direkterekyl* (F8b+), Bergflødt, Norway. The combination of small handholds and footholds, where most of them are either sidepulls or gastons, makes this route a proper testpiece for balance and weight distribution.



WEIGHT DISTRIBUTION

Pull on with two not-too-good handholds on the wall, and use two footholds which are spaced slightly further apart than the handholds. Move your right hand to a hold further up to the right and back down again, then move your left hand to a hold up to the left and back again. Before moving your hand, move your body from side to side to find the position with the best balance. Start with larger movements and gradually make them smaller as you start to find the balance point.



Get your body used to finding a sense of balance. This is an important skill you'll need throughout your climbing career. Even experienced climbers depend on this every time they're facing challenging and technical moves on small holds. It might look like they're 'just' climbing, but even they are forced to use their body to get a feel for what's right for them.



TRAVERSING

Traversing is a simple yet effective exercise for training balance. When moving sideways, you're forced to constantly shift your centre of gravity to maintain balance. Mainly using sidepulls will add to the need for finding the right body positions. Add in a bit of traversing as a part of every warm-up, and focus on climbing in control and in balance.



Choose walls with lots of holds, which will allow you to use larger footholds and smaller handholds.

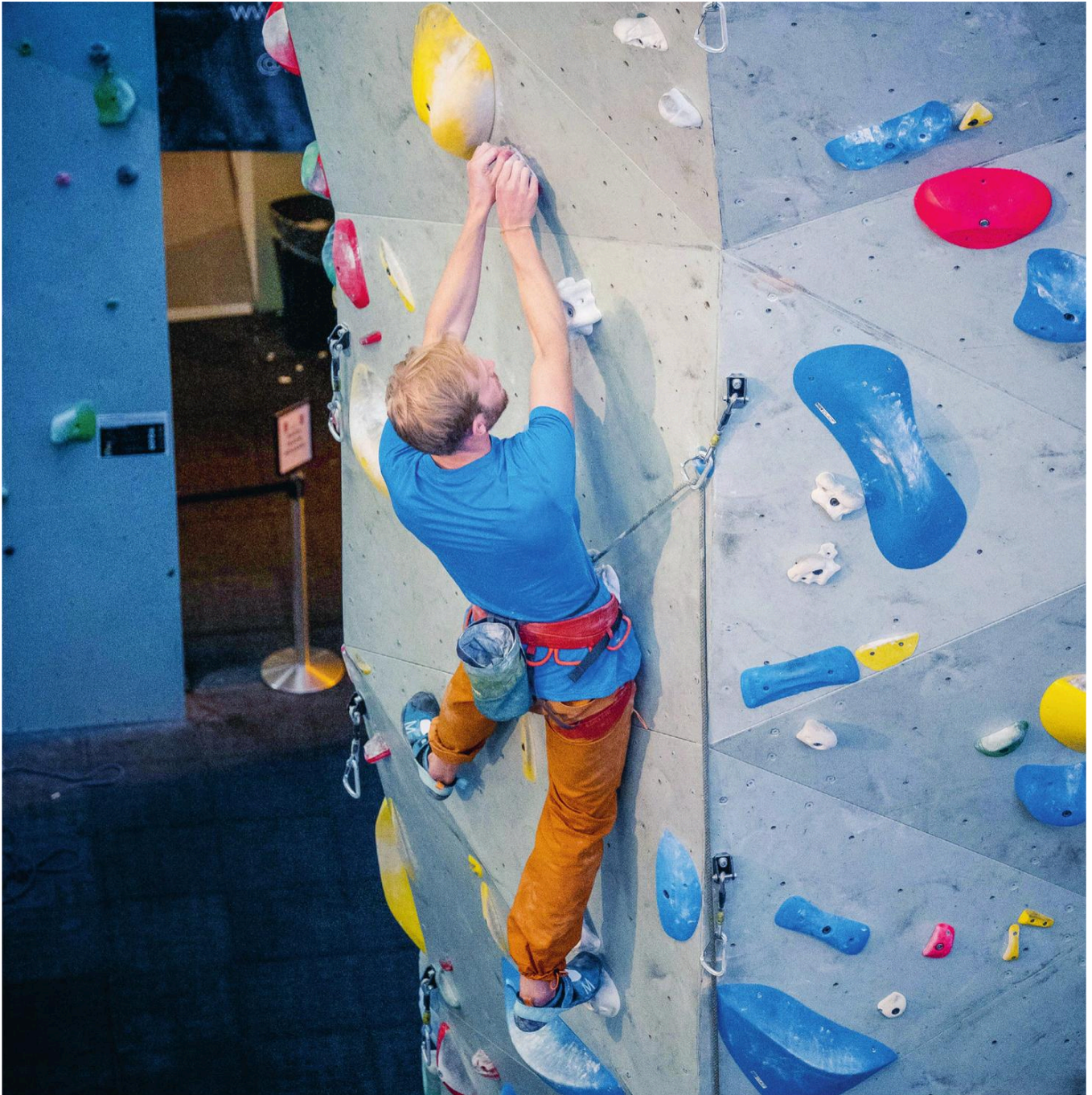


LOCKING OFF

If you are forced to lock off in order to grip a tiny hold where you really have to place every digit perfectly on to vanishingly small features, you will be dependent on finding the perfect balance so as not to miss the hold or waste energy. The same goes for clipping the draws when climbing on lead. A good exercise for training balance is therefore to climb easy routes but lock off for 1 to 3 seconds each time before grabbing the next hold. Here it's important that you find the optimal balance before you move your hand to the next hold. To make the exercise more difficult you can climb progressively harder routes.



Climb on vertical routes and take your time!

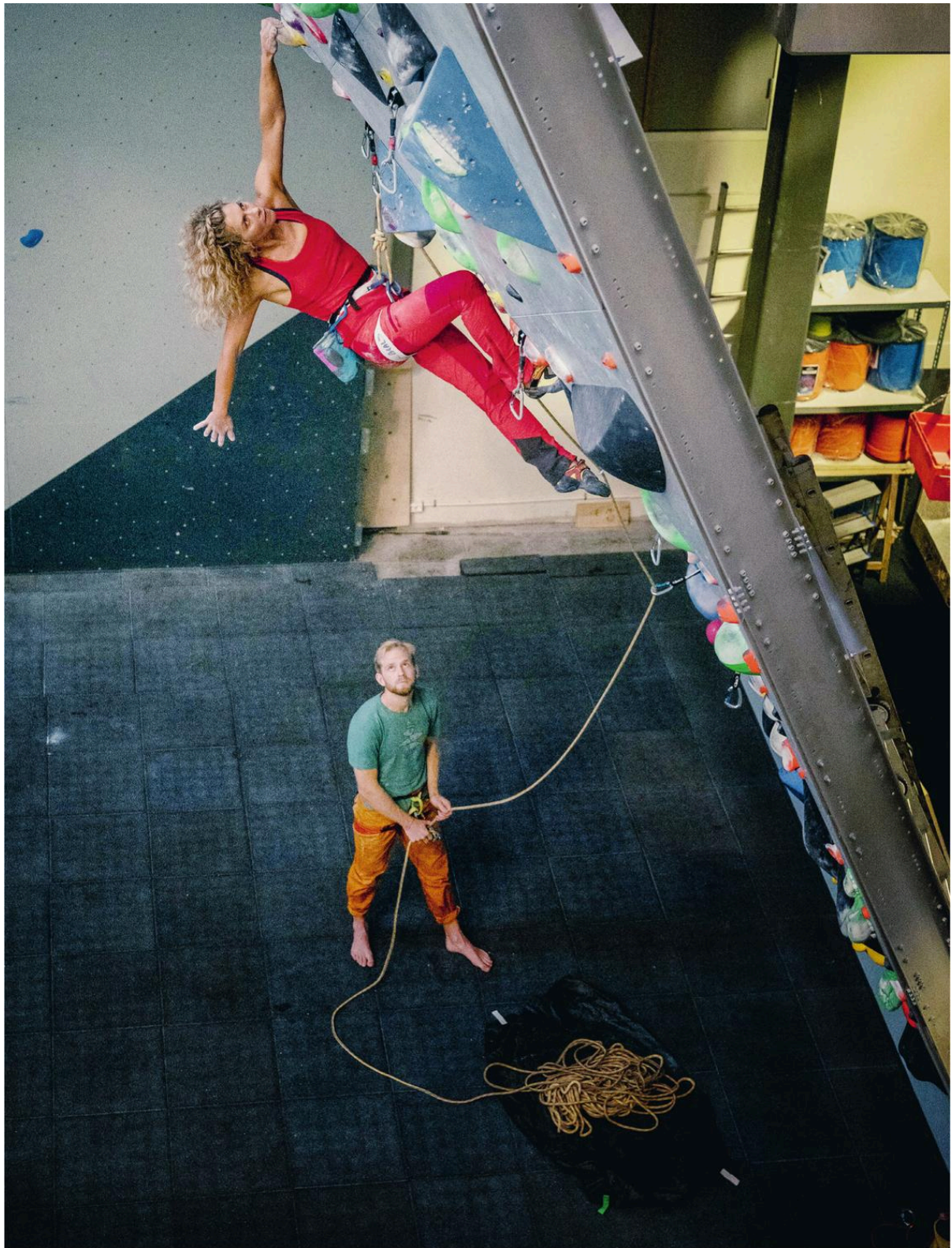


MATCHING

If you have to match a hold, you're dependent on finding the right body position such that you're in balance when matching – especially if the hold you're matching is a sloper. You'll also have to lower your body, in order to get a better angle in relation to the hold. For this exercise you should climb routes well within your limit and match every possible hold as you climb. When you've climbed several routes by matching all the holds you can play with harder routes and find which holds feel natural to match. It's often advantageous to match holds if you're about to do a long move and if the hold is large enough to fit both your hands. It can also be smart to match hands before a high foot placement, but climbing is a sport with many exceptions, so you'll have to get a feel for it yourself.

NO MATCHING

'No matching' is well known in the climbing wall, but does this kind of training make us forget to match when we're supposed to perform? Regardless of the answer, it's important to also train matching to become familiar with the advantages it can yield on the wall.



FIND THE REST

Finding good resting positions on a route is an excellent way to train your balance. In order to be able to rest you have to lower your body, get as much weight on to your feet as possible and, most important, find your balance. If you're not in balance you will have to use unnecessary amounts of energy to stay in place and you won't be able to rest properly. We suggest trying to climb vertical routes two to three grades below your maximum, and trying to find five rests where you can hang for 20 to 30 seconds. It's important that you can breathe calmly, shake your hands and chalk up if you need to.



If you can comfortably hang with a straight arm on an open-handed grip, you've probably found a good resting position.

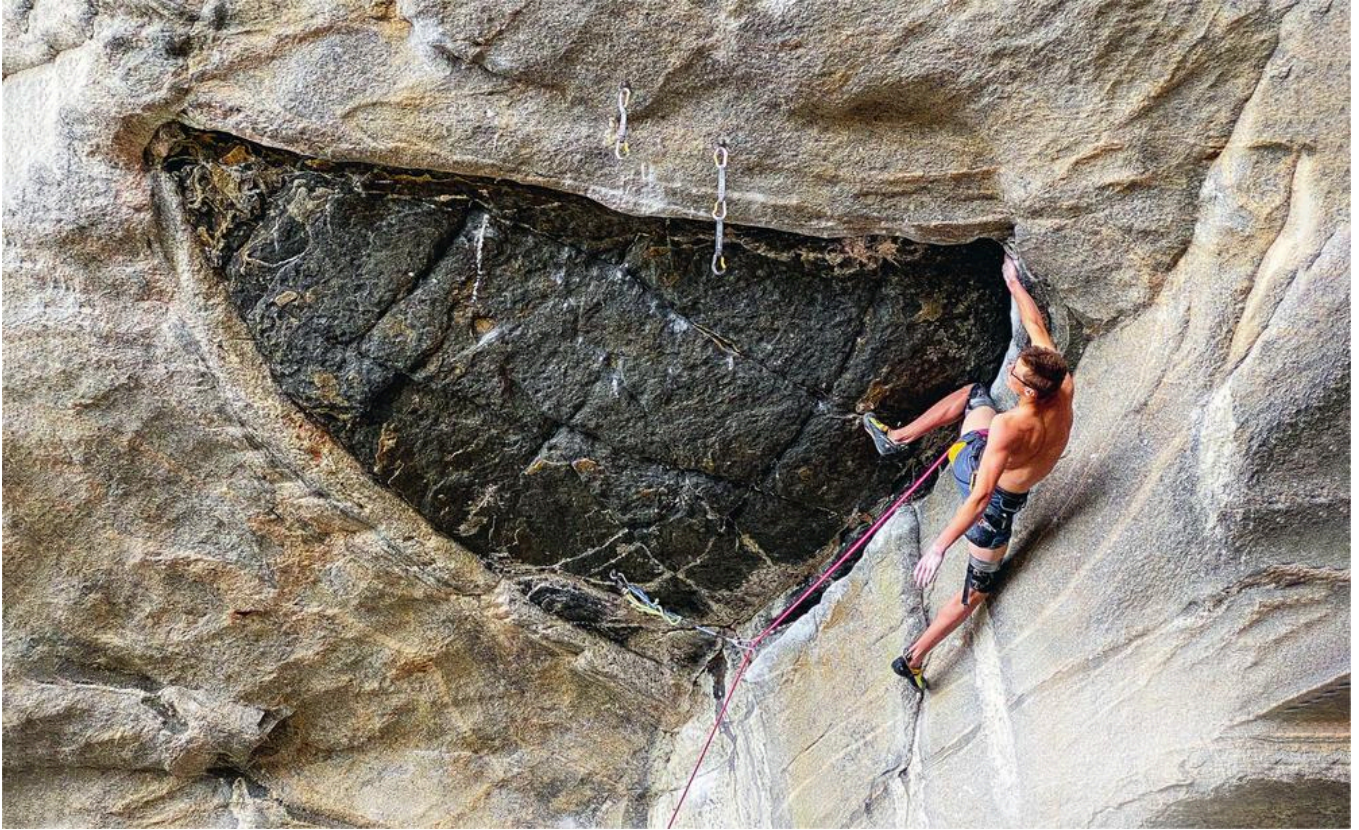


PHOTO: MARTIN MOBRÅTEN

Outdoor routes are often comprised of hard sequences followed by good rests. Learning how to rest while climbing is therefore an important skill to master if you want to climb outside. Here, Arne Farestveit is relaxing before the crux of *The Eye of Odin* (F8c+), Flatanger, Norway.



CLIMBING WITHOUT PINCHING

It's better to avoid pinching when training balance. We sometimes call pinching the 'technique killer', because in many cases it removes the need for proper technique, as long as you're strong enough to pinch a hold. Pinching enables you to lean your body out from the wall and use your arms and hands to pull yourself in. This allows you to climb without having your body in the right position. Let's use a pinchable sidepull as an example:

If you pinch the hold you'll be able to lean out from the wall. If, on the other hand, you're only gripping the hold using an open hand you're forced to position your body to the side and in towards the wall in order to hang on.

Start by trying boulder problems or routes you know well and try to repeat them without pinching. Then you can move on to trying new routes in the same fashion.



CLIMBING WITH FEWER FINGERS

By adding limitations to how you grip the holds you'll automatically transfer more weight on to your feet. A hold which feels good can lead to you pulling too hard with your arms and being careless with the little details that decide whether or not you're in balance. By using only two or three fingers on each hold you'll reduce the chances of making these mistakes. You should perform this exercise on slabs or vertical walls so as not to put too much load on your fingers.



Try a boulder problem using your preferred grip position. Then try climbing it with only three fingers on each hold, and then just two fingers. How does this affect your climbing?



ONE HAND ONLY

To make things even more challenging, try climbing a few boulder problems using only one hand. For this exercise it's important that you choose vertical walls or slabs, and problems with relatively large footholds. By taking the time to find your balance before moving your hand to the next hold you can maintain full control while climbing.



Find your balance, trust your feet and do short moves.



NO HANDS

As a final challenge, try to climb without using any hands at all. You'll need to find boulder problems on slabs with large holds or volumes for standing on. Start by using all available footholds before progressing to single problems. Make smaller steps to avoid losing your balance.



To make sure this is a balance exercise, try to move as calmly as possible to maintain balance throughout the movement. If it's not possible to do a certain move statically and in control, try to generate as little speed as you can.



PHOTO: KNUT LERHOL

Martin Mobråten trying to climb without using his hands.

JOHNNY'S METHOD

British climber Johnny Dawes is well known for his impeccable technique and, at that time, unconventional training methods. He used to combine slabs and dynamics in ways that were almost unheard of. Outdoors, and especially on slabs, you were supposed to climb in a controlled way and not move dynamically between often poor holds. About 20 years ago Johnny was holding a climbing clinic at the Tyrili Climbing Centre in Oslo. We took part in this clinic and to be quite honest, we didn't think it was any good. We were mostly instructed to climb with our hands on our backs, and only on the really difficult problems were we allowed to perhaps use just one finger. Come on! We wanted to climb hard and we realised this was never going to make us any stronger. We left the clinic disappointed, without feeling neither better nor smarter. But here we are, 20 years later, and we've written a book filled with exercises that are eerily similar to what Johnny tried to teach us. Johnny believed you had to challenge your familiar repertoire of movements, limit the way you could do moves and try new things all the time. After a couple of decades we've come to realise the same thing. Not all the time of course; there's much to be gained from training basic skills and we should be training 'regular' climbing as well, but that's basically what we do on the wall all the time anyway. That's why it's so important to challenge ourselves with new exercises and new variants of moves we've done before. We hope this story and this book inspires you to try something new.

TENSION & DIRECTION OF FORCE

IN ORDER TO achieve tension and the correct direction of force you need to position yourself in such a way that you can use the holds in the best way possible. This is essential when climbing on sidepulls and undercuts. Just imagine how difficult it is to use a right-hand sidepull if there's no foothold further out to the right that allows you to weight the sidepull. Similarly, you can use opposite-facing surfaces to create tension between the holds. For example, imagine climbing a prow with slopers on both sides. The holds by themselves might be terrible, but because you can compress between them they might even feel rather good.

The exercises to train tension and direction of force resemble the exercises for training balance, but they are mostly performed on steeper walls. The central point here is mostly about positioning in relation to the holds and in which direction to push or pull, and not solely about making fine adjustments to your centre of gravity. The following pages feature suggested exercises.

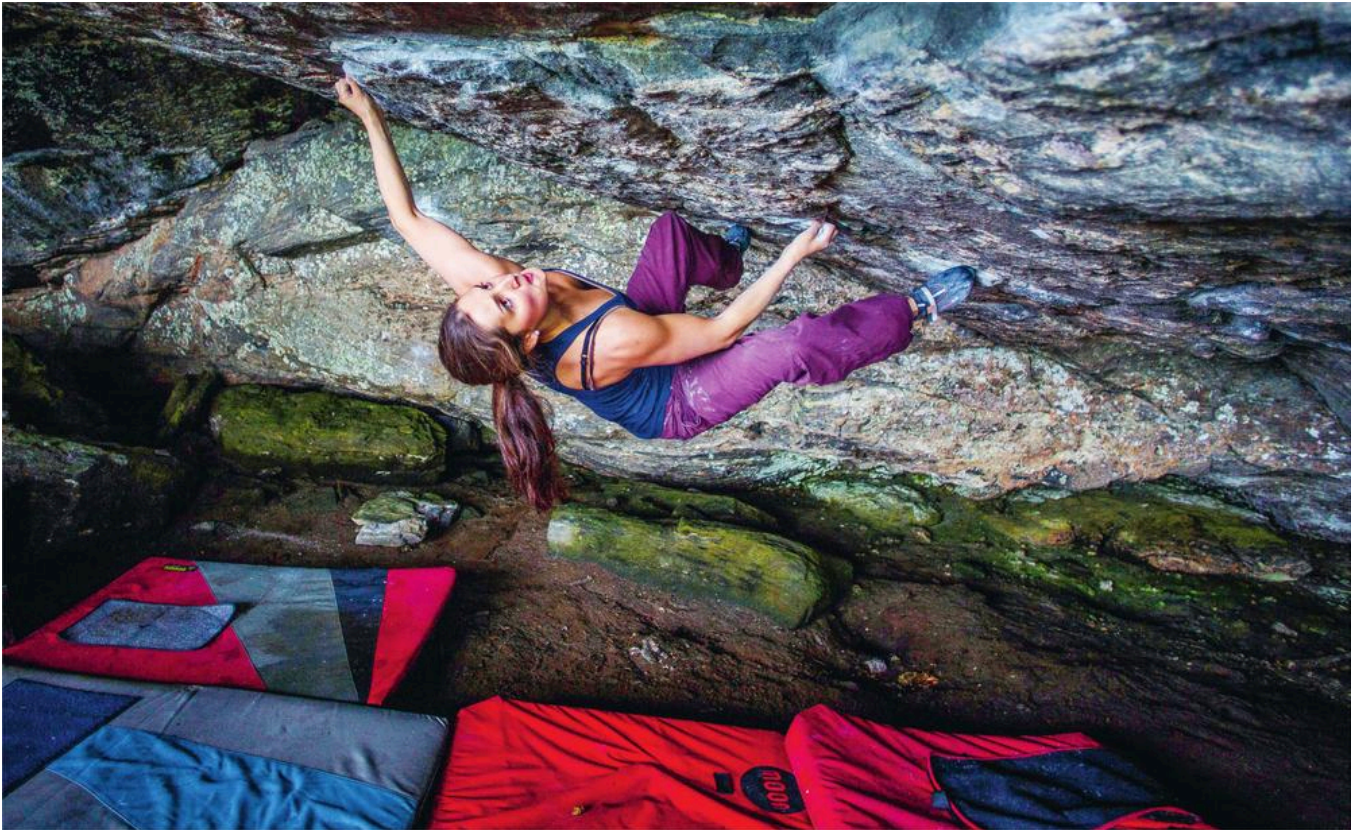


PHOTO: ENDRE VIK

Maria Davies Sandbu tensioning up on *Wolverine* (Font 8b+), Vingsand, Norway.



FINDING THE POSITION

To train yourself to get into the correct position to execute moves, try climbing routes where you have full control of every move. Don't take shortcuts by moving dynamically or kipping past tricky sections.* You should position yourself in such a way that you can do every move statically, or close to it.

* 'Kipping' means to swing dynamically through a move, either because of necessity or the inability to do the move in a controlled fashion.



SIDEPULLS ONLY

For this exercise, as far as it's possible, try to climb only on sidepulls. You should choose a wall with lots of holds so that you have a good selection of sidepulls when climbing. Don't worry about the colour of the holds or the intentions of the route setters. Focus on finding the right body position for every move to stay in control when moving up the wall. It's important to climb on relatively easy routes so that you have enough strength to complete the exercise in good style.



When you have completed this exercise a few times, try doing it without pinching the holds. This will require even better body positioning.



CROSSING THE LINE

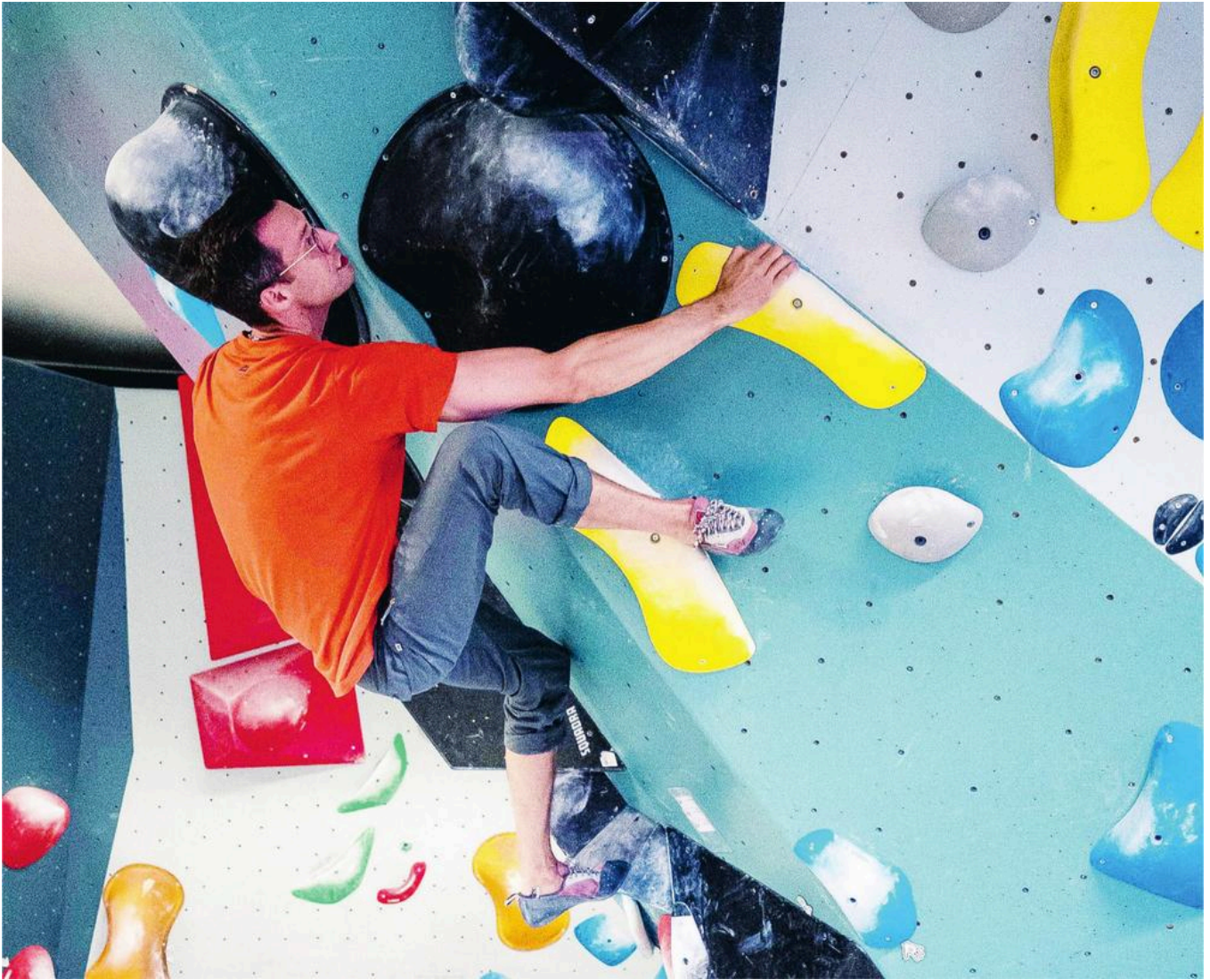
For this exercise, start on two good handholds on the bouldering wall. The wall should be vertical or slightly overhanging, and you can use any footholds. Now cross your left hand past your centre line, so that both of your hands are on the same side of your centre line. After completing the cross move, position your body so that you're ready to release your right hand and move it to a hold that's further to the right. Repeat the exercise with your right hand. Now you can play around and see how far you can do each cross move. As the cross moves start to get longer, it'll become natural for you to turn your left hip into the wall when crossing with your left hand, and vice versa.



Try crossing from a sidepull to a sidepull that faces in the opposite direction. How does this affect your body position as you prepare to do the next move?

THE ADVANTAGES OF TWISTING: MARTIN SHARES A STORY

I started climbing over 20 years ago at Møt Veggan ['Meet the Wall', Ed.], a climbing wall in Skøyen, Oslo. There was a climber there named Markus. Markus wasn't particularly muscular nor was he very strong, but he moved like a cat on the wall. I found this fascinating and I would just sit and observe Markus climbing. What I noticed was that he twisted in on a lot of the moves. By using this technique he was able to climb hard grades without using a lot of strength. This was something I had to try. For a while, I was trying to twist in on every move. After some time I realised that this wasn't always helping, but I had still learned a technique that quite often proved itself useful. Using this technique removed a lot of the need for power when just fronting hard moves, and I was able to do the moves more easily as I positioned myself from side to side by twisting in.



COMPRESSION

To train compression we want you to climb prows and test out different body positions. Use any holds for hands and feet, and try to find a prow with a lot of holds for this exercise, so that you have many options to choose between. Try climbing the prow with a ‘T-rex’ compression, a wide compression and a nearly fully stretched-out compression.[‡] Start with footholds that place you in a position with your arms at shoulder height, then try using footholds that are slightly too low and then slightly too high, because then you’ll have to compress with your arms either above or below your shoulders.

How strong you are at compressing in different positions can vary greatly. As a general rule, you’re strongest when your hands are placed relatively wide and at shoulder height. It’s also advantageous if your right and left hands are placed at equal distances from your right and left shoulders, respectively. In other words, you should strive for symmetry and not offset positions when compressing.[‡]

[‡] We call it T-rex compression when both handholds are near one another. The closer they are to your centre line, the more your arms will look like those of a Tyrannosaurus rex. This position offers less than ideal working conditions for the muscles in your chest, meaning you’ll be weaker in this position than if the handholds were placed further apart and you were compressing wider.

[‡] Offset means that the holds are at different heights in relation to one another.

T-REX

You’re not strong in the T-rex position, so try to avoid it if you can.



PHOTO: MARTIN MOBRÅTEN

Maria Davies Sandbu compressing between the holds to send *More than I Had* (Font 8a), Flatanger, Norway.

DYNAMICS

MOVING DYNAMICALLY IS both a technical and mental challenge. Technically, you want to lower your body so that most of the movement is generated by the larger muscles in your legs, and then use your arms to guide your momentum towards the next hold. As you reach the next hold with your hand you'll want your body close to the

wall and at the apex of its upward trajectory – the point where your body stops to move before it begins to fall back down again. In order to achieve this, you will need to allow your body to move away from the wall slightly at the start of the movement, and then pull it in towards the wall as you move up. You also need to have the correct timing, so that you have just enough speed to reach the next hold with your body in the position you want. For many, it's a mental challenge to let go and move fast for the next hold. This mental barrier will hinder your progression as a climber, because the benefits of moving dynamically are great. You can do bigger and harder moves, and you'll save energy by swinging between the holds instead of locking off.

In our experience, it's best to start training dynamic climbing by doing exercises that don't require lots of speed and coordination. As long as the walls aren't too steep you should be able to perform the moves quite slowly and you'll have plenty of time before you have to grab the next hold. This will allow you to focus on the technical execution and build confidence on these types of moves. By starting easy you'll start to believe that these moves are possible – even for you.

DYNAMIC MOVES

A dynamic move is not the same as a dyno. Using speed to reach the next hold is not the same as jumping for it.



PHOTO: MARTIN MOBRÅTEN

Stian Christophersen swinging his way up *Diamanten* (Font 8b), Vingsand, Norway.



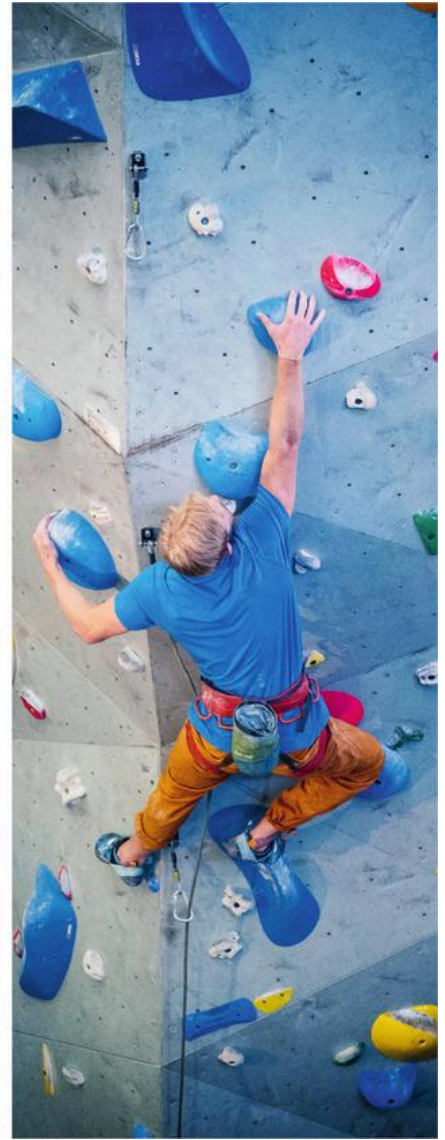
WALL BARS

Wall bars are a great place to start your dynamic training. You can easily find the level that's right for you, and focus on correct technical execution and gain confidence in doing dynamic moves. Here's a progression ladder you can use:

- Start by holding the same bar with both hands and stand on a bar that has you bending your knees at 90 degrees or more when your arms are straight. Push with your legs and shoot upwards, while at the same time pulling yourself in towards the wall with your arms. Move one arm up to the highest bar you can reach. Grab the bar with a slight bend in the elbow. Switch arms and repeat. To make the exercise harder you can place your feet higher, but still start with your hands on the same bar as before. This requires more strength in your arms to pull your body in towards the wall – the movement will be faster, and it will be harder to get the right timing. Continue repeating the exercise with both hands until you get the timing right, meaning you are grabbing the top bar with your hips against the wall and your body at the apex of its upward movement.
- Repeat the exercise, but this time move both hands to the highest bar you can reach. This is more challenging mentally, as you have to let go of the bars completely, but as long as your feet aren't too high, you'll notice that you actually have quite a bit of time. If you think it's too scary you can start by going for the next bar above your starting bar, and then increase the distance. As you get more comfortable with the exercise, try clapping your hands before grabbing the next bar. With a bit of practice, you'll notice you can clap one, two or even three times before grabbing the next bar. As with the previous exercise, repeat this exercise over and over until your timing is perfect.
- To make this exercise even harder still, try also moving sideways in addition to upwards. This time you should grab the next bar either to the right or the left of your starting position. You'll have to lower your body to generate speed upwards, but also swing from side to side to create the correct arch towards the bar. This requires even better timing, but is also more relevant to climbing. Again, repeat the exercise until your timing is perfect.



If you don't have access to wall bars you can also do this exercise on a vertical wall with a good selection of jugs, so that you can easily adjust the difficulty.

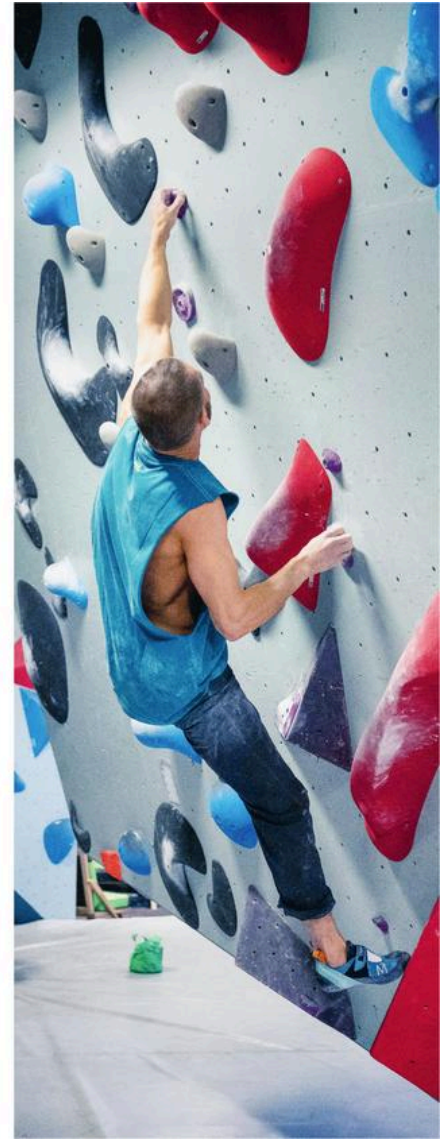


DOWN AND UP

For this exercise, choose a route you know well. The wall should be slightly overhanging and the holds should be quite good. Lower your body and start every move with your arms straight. Initiate the upward movement with your legs and pull your body towards the wall with your arms. Try to deadpoint the moves and grab the next hold with your elbow slightly bent, so that you're able to lock off your body as you stick the hold.



Try to flow through the movement and avoid unnatural changes in direction or stopping after the movement has been initiated.



DEADLY PRECISION

Choose two good holds on a vertical or slightly overhanging wall and choose footholds at a comfortable height. The goal here is to move dynamically to a hold that's higher up than you are able to reach statically, but not higher than you're able to reach after trying a few times. When you feel you're in control of the move, repeat it at least 10 more times. Even if you can do the move first go, it's far from certain that you're executing it perfectly, and the goal of this exercise is to perfect the movement. A good indicator of how well you're executing the move is how much energy you're spending to stop the movement as you grab the hold you're going for. In theory, you shouldn't be using any more force to stick the hold than if you were to do the move statically, but this won't be the case if your hips are too far away from the wall and your body is still in motion. Therefore, you need to actively work on these key elements on every try. Think about how the move feels, video yourself and get feedback from others. When you feel you have optimised the move, switch hands and repeat the exercise.



Cecilie using a sloper and a guide hold.

To make the exercise harder, you can add rules for how you're allowed to grip the holds. It'll be significantly harder if you're only allowed to grab even relatively good holds with just three or two fingers. Don't try to grab holds with just two fingers before you've practised the move, as too much force on your fingers as you stick the move can lead to strained muscles in your forearm. The next step is to choose holds that require a bit more aim, for example pockets, blocked holds or slopers. These will require a higher degree of precision in your movement. Repeat the different variants 10 times each and search for the perfect execution. As a last challenge we want you to make the starting holds worse.

Here are some suggestions for challenging starting positions:

- Small, positive holds allow you to drop your hips away from the wall slightly to create speed in towards the wall as you reach the next hold. It's likely that you will have to start moving in towards the wall sooner than if you were starting on jugs. It will be harder for your fingers and it will require a higher degree of precision in your movement.
- Use a sloper with your upper hand and a better hold – a guide hold – for your lower hand. A positive sidepull or an undercut is preferable for your lower hand, so that you can compress between the holds. This will allow you to drop your hips away from the wall a bit to create momentum in the right direction in order to reach the next hold with the correct body position. You should be grabbing the next hold with your lower hand from the guide hold, and not the hand which is on the sloper.
- Using two opposing slopers is even more challenging, but if you can compress between them you can create the right momentum towards the next hold.

REPETITION

If you find it boring to repeat the same move 10 times, consider the fact that the Japanese competition climbers, arguably the best team in the world at the moment, often play around on the same move for up to an hour before moving on to the next part of their programme.

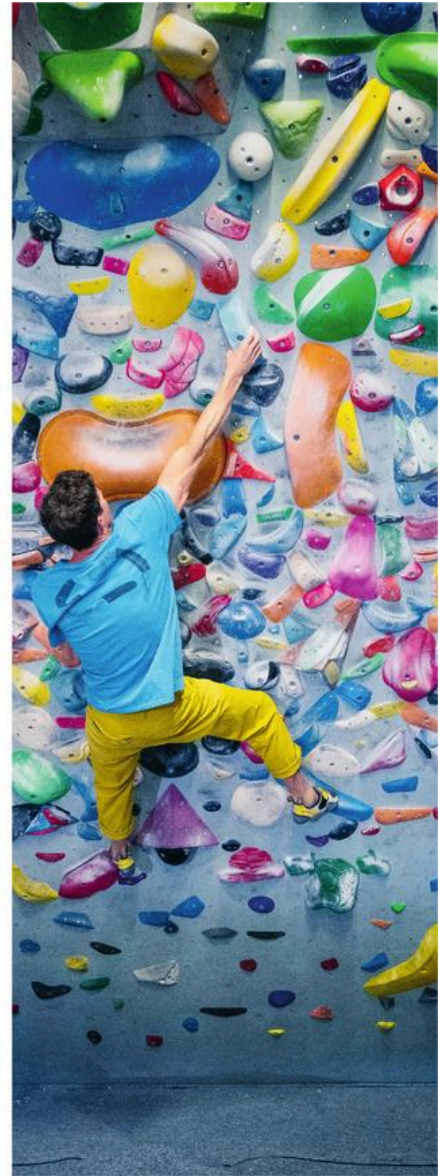
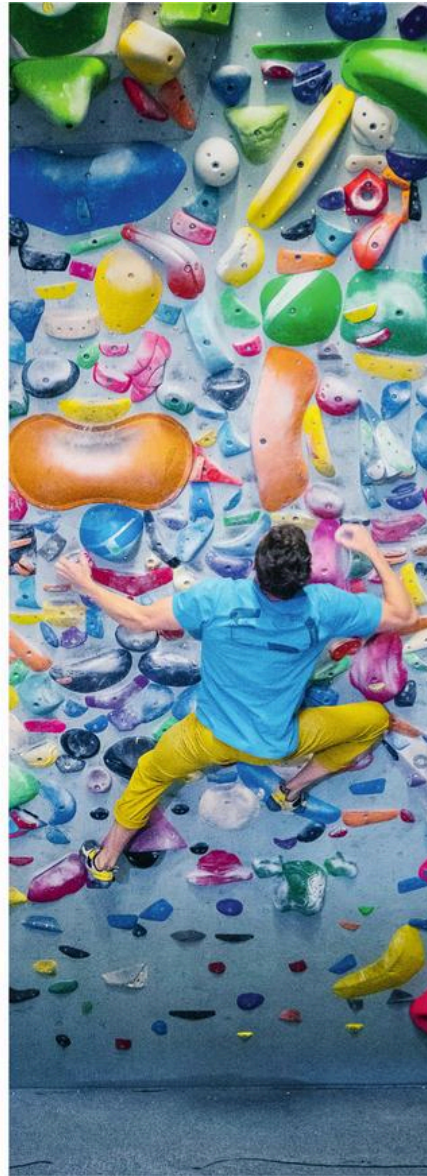
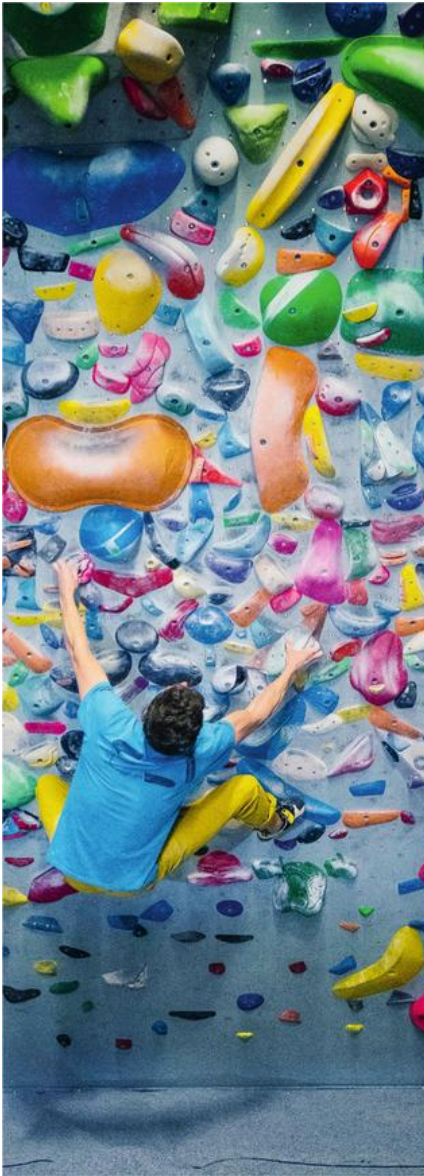


DYNOS

We don't want to give too many pointers for your dyno training. Simply put, you should be playing around with as many different dynos as possible. Start with simple dynos between good holds on vertical walls. Do both regular and double dynos, so dyno with just one hand or both hands, and gradually increase the lengths of the dynos and the steepness of the wall as you gain confidence with these moves. It's usually advantageous to overdo the speed a bit, so that you land with your elbows bent. This way you'll be able to quickly pull up to avoid an unnecessarily large swing. Focus on locking off your body as much as possible as you stick the hold.

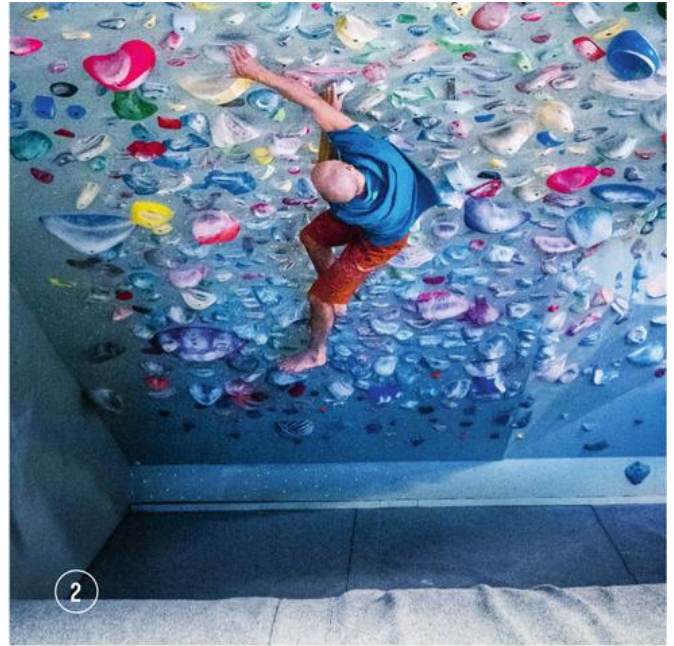
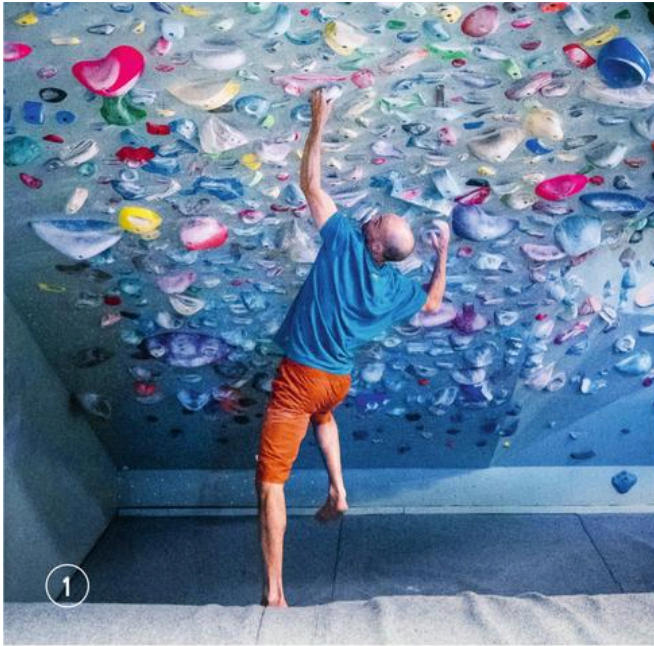


Always incorporate double dynos at the end of your specific warm-up routine. This activates the muscles and helps you to loosen up mentally.



ARCHING

Doing a sideways arch instead of moving straight up towards the next hold can be especially advantageous when climbing on sidepulls. To train these types of moves, find a layback edge or use the side of a large volume. Grab the edge with one hand and find a hold lower down with your other hand, as with the exercise with the sloper and the guide hold [mentioned earlier](#). The footholds should be relatively high, so that you're able to do a long move to the next handhold. Now, if you try to move straight up to the next hold, your centre of gravity won't be far enough to the side and the direction of force will be all wrong. You will therefore have to start the movement by shifting your centre of gravity to the side and away from the sidepull, before pushing up with your legs to create upward momentum.



CROSS MOVES

When doing dynamic cross moves where you cut your feet from the wall, you should open your hips and front the wall instead of twisting in. This is especially relevant when the wall is overhanging. If twisting in, your body would continue to rotate after completing the movement, which might make it difficult to hold the swing. If you front the wall instead you'll avoid this rotation, because your body's initial relative position is the same as when you land the next hold.

To begin with we recommend that you stand on the ground below a steep wall (preferably 40 to 60 degrees overhanging) and try to jump to a hold with one hand while holding a handhold lower down with the other. It should be a cross move, so if you are to jump with your right hand, the hold you are aiming for should be further to the left than the hold you are jumping from. First, try to jump while facing towards the hold you're jumping to, then turn and jump while facing the wall.

Vary the length of the cross move and notice the effect it has on your rotation and how you should position yourself before you jump. If you really overdo it, you might have to start with your back against the hold you're jumping to ([see photos](#)). It will feel like you're jumping backwards, but even though this might feel wrong, it's still correct.

The next step is to do this on the wall. As long as the holds are good enough you can use the same holds as before. And as before, start by twisting your hips in and position yourself as if you were to do the move statically,

but try to jump to the hold. Then choose footholds that force you to open your hips and front the wall – there's no twisting in this time. Vary the length of the cross move and see what happens.



TRICKS

MODERN PROBLEMS REQUIRE modern solutions which you'll need to master if you want to climb hard problems indoors or participate in bouldering competitions. Techniques and styles like 'flicking', 'clutch' and 'wall runs' are commonplace and require a great deal of coordination. You should therefore regularly practise these techniques in order to master them.

Luckily, they usually look the same on the wall, with the holds arranged in the same way, so you can practise these techniques without the boulder problem being set in this exact way.

It's easy to forget the fundamental elements when practising more advanced moves. Therefore, always remember that dynamic moves should start away from and move in towards the wall, and that the movement should be as close to a deadpoint as possible. Also remember which compression position you're the strongest in, as flicks and clutch moves require compression strength.



PHOTO: BJØRN HELGE RØNNING

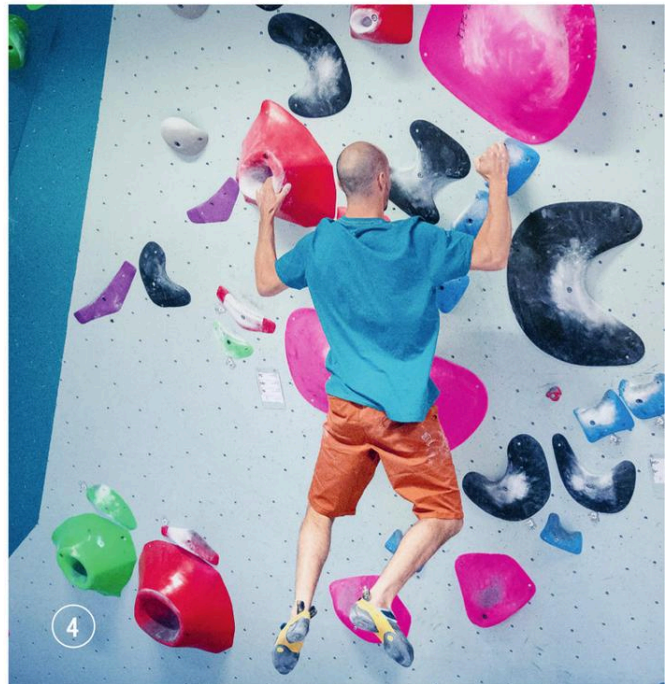
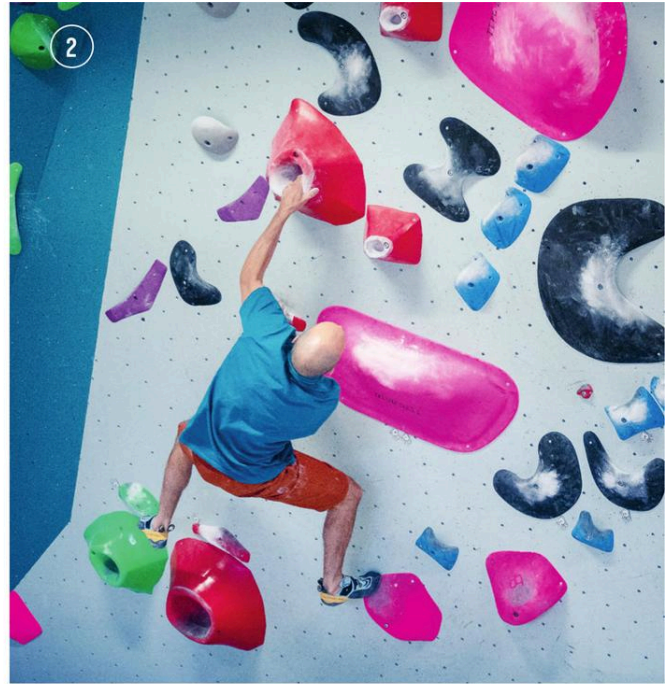
Stian Christophersen on the last problem in the final of the 2017 Norwegian National Bouldering Championships. At the start of the boulder problem you had to do a dyno while at the same time flicking your right foot into a toe hook. From there you had to do a double clutch out of this position to top the problem.



Focus on the first hold you're going for and let your other hand find the other hold naturally.

FLICK

A flick is when you move one hand to a hold and, either in the same movement or immediately after, have to turn – flick – the other hand to another hold in order to maintain your position. The hold arrangement is usually similar to what is shown in the [photo](#), with sidepulls all facing the same way and the hold you're flicking to facing the other way. Start by practising on vertical walls with very good holds and short moves, where you don't really have to flick at all: the idea is to practise the technique and get used to it on easier climbs. After mastering the technique on good holds, you can progress to smaller and more slopy holds, and then increase the distance between them and choose steeper walls.



Look at the first hold you're going for and let your other hand swing naturally to the other hold.

CROSS CLUTCH

A cross clutch is when you're do a cross move and just as you grab the first hold you continue with your other hand to the next hold in order to achieve a better position. The hold arrangement is almost always a cross to a sidepull and then paddling to another but opposing sidepull. Start practising on slightly overhanging walls and good holds that are ever-so-slightly positioned as sidepulls. Do a short cross move, and just as your hand lands the hold, swing to the next hold that's a bit further away. After a while you can progress to bigger moves, steeper walls and poorer holds.



DOUBLE CLUTCH

The double clutch is a continuation of the double dyno, where the first hold you land isn't good enough for you to be able to hold the swing, but instead is used to generate more speed or change direction towards the next – and hopefully last – hold. The arrangement of holds is usually a dyno to a sloper, or a similarly bad hold, then continuing on to a good hold. We recommend starting by dynoing to a jug and then continuing on to another jug immediately above. After a while you can play around by increasing the distance between the holds, adding sideways movements and using poorer holds. We recommend doing this exercise on walls that are around 20 to 40 degrees overhanging.

How much of the swing you should try to control on the first hold before moving on to the next can vary a lot. Generally, it's better to move quickly if the first hold is bad and the next hold is good. If, on the other hand, the first hold is good, it's better to try to stop the swing as much as possible before moving on to the next hold. We recommend that you play around so you that you get a feel for what's best.

INDUSTRY OF COOL

There's a boulder problem in Rocklands, South Africa, called *Industry of Cool*. The crux of this boulder is a double dyno around a steep prow to a sloper that's so bad you need to make a clutch move with your right hand into a good hold in a crack. Because it's hard to hold the swing on the sloper, it's natural to pull up and move on to the next hold quite quickly, which is exactly what Thilo Schröter and I (Martin), were attempting, again and again, some time in 2017. We felt we were really close, but we were never able to stick the good hold in the crack. I'm pretty sure I've tried that move over a hundred times spread out over several days and trips to the area. And always with the same result: I'm unable to hold the swing long enough to reach the next good hold. After watching a video of American climber Dave Graham sending the problem, we returned a few days later with a new tactic. In the video, Dave takes most of the swing with his arms straight before, at the end of the swing, he elegantly and frighteningly easily pulls into the crack. After just a few attempts, Thilo managed to stick the move and quickly found himself on top of the boulder. It still wasn't easy, but the new tactic worked. This just shows that the theory of pulling up early on bad holds isn't always correct. Sometimes it's about muscle power and pulling up, other times it's about 'skeleton style' and straight arms.

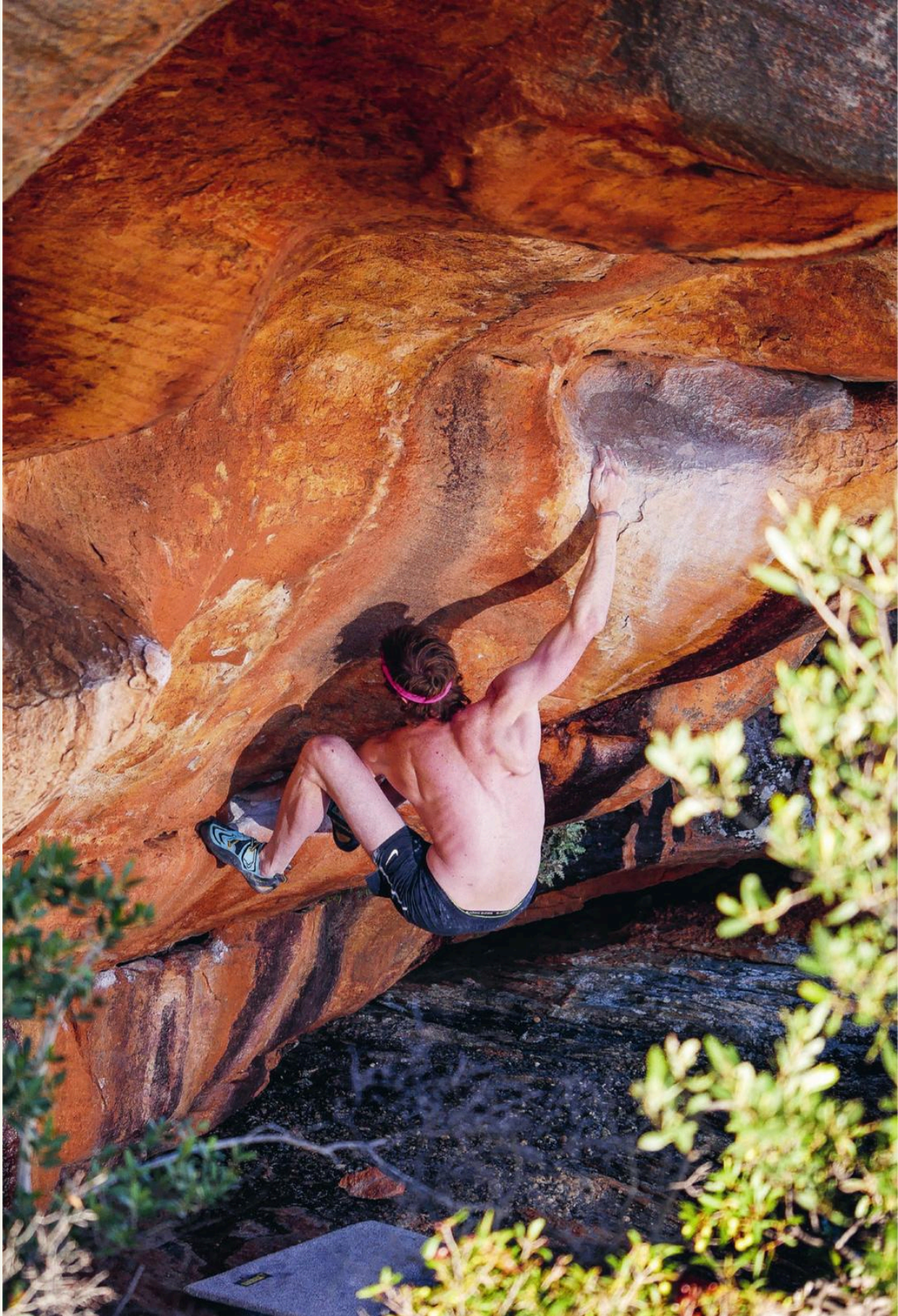


PHOTO: MIKE MULLINS

Bjørn Sætnan on *Industry of Cool* (Font 8a+). Bjørn is taller than most and found his own way of sticking the sloper – by not letting go of the starting hold. This is one of the things that's so cool about climbing – you can find your own way up, using the resources you have. It's still really hard the way Bjørn climbed it, but his post on Instagram was hashtagged *#cruxesareforshortpeople* ...



Try standing far out on the starting volume as this makes it easier to generate momentum in towards the wall.

WALL RUNS

Wall runs are a popular addition to competition climbing because they make for spectacular viewing. They entail running/jumping on holds, usually volumes, without any handholds to grab on to along the way. To start practising wall runs all you need is a few volumes low to the ground and a selection of good handholds. Close-to-vertical walls are best, but slabs and slightly overhanging walls will do just fine too. To begin with, play around with short moves where there's no real need to run at all. Maybe all you need is a step to the side in order to reach the next hold. Then you can progress by increasing the distances between the holds – this will require more speed and better coordination. Also play around with sidepulls and poorer holds.

THINK OUTSIDE THE BOX

Before the 2018 Norwegian Bouldering Championships in Trondheim, several of the local athletes paid multiple visits to Rush Trampoline Park in order to improve their dynamic abilities from a technical and mental standpoint. Just think about it – a small deadpoint move on the wall will be less challenging if you did a backflip on the trampoline the day before.

COMBINATION EXERCISES

CLIMBING IS TECHNICALLY complex, and you constantly have to solve new problems while climbing. You have to deal with delicate footwork while at the same time trying to maintain balance, in addition to having to choose between using speed or locking off for the next move. The exercises we have presented so far more or less focus on isolated elements, and it's important to be able to master each element separately. But while climbing it's also important to keep all of these elements in your mind so you are quickly able to choose the most appropriate technique for the situation and to combine the different elements, if necessary. Here's a selection of exercises to help you practise this.

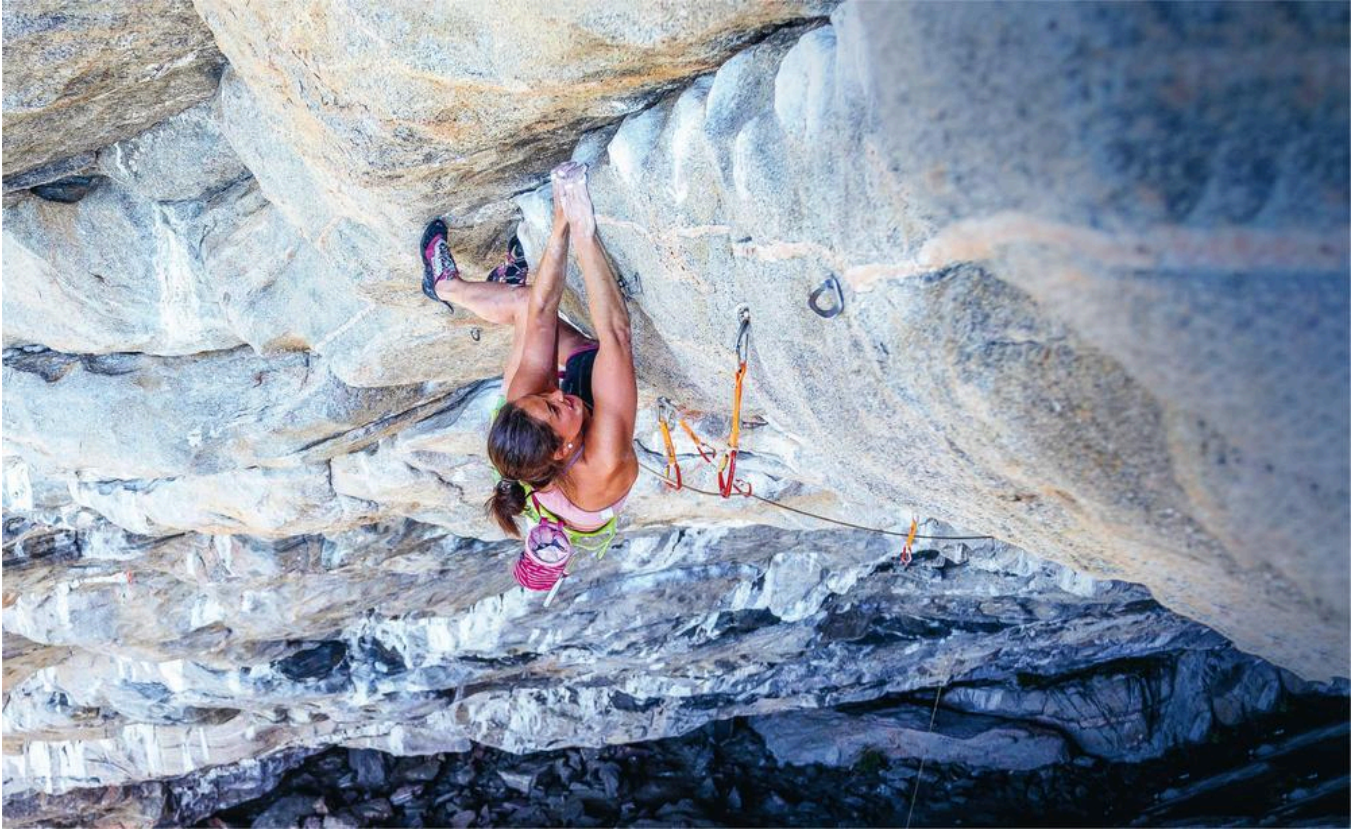


PHOTO: HENNING WANG

Maria Davies Sandbu fighting her way up the last rail on *Nordic Plumber* (F8c), Flatanger, Norway.

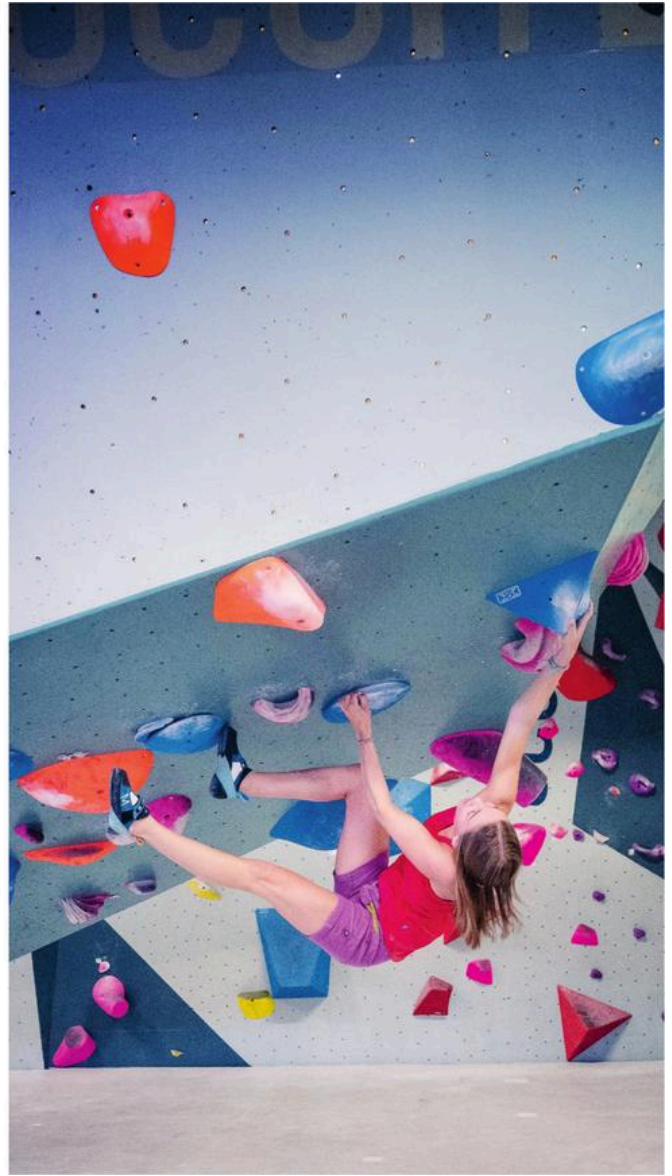


STATIC **VS** DYNAMIC

Part of what makes climbing difficult is figuring out how to attack a move. Should you take your time – finding the perfect balance – and execute the move statically and in full control, or should you swing through the move and do it dynamically? A good exercise for practising these choices is to climb a selection of boulder problems by first doing all the moves statically, and then doing them dynamically. After completing all of the problems twice, try to combine the moves in the best way possible. Which moves were easier to do statically, and which moves were easier to do dynamically? Be honest with yourself. If you're a static climber you have to be conscious of this and not think that all the moves were easier to do statically. If you find this difficult, you can discuss it with a friend who has different qualities as a climber to yourself. We would argue that it's always good to climb with people who have other strengths than yourself, so that you can learn the best parts of their climbing style and add these to your own.



Start with easy boulder problems on slabs and vertical walls, to make sure you'll be able to do all of the moves statically.



5 PROBLEMS IN 5 MINUTES

To become better at adapting to different styles you should change between different wall angles and holds when climbing. A good exercise for training this skill is to climb five different boulder problems within 5 minutes. Choose one problem for each wall angle: slab, vertical, slightly overhanging, roughly 45-degree overhanging and roof. It's good if these are problems you've tried before so that you know what to do and you know you'll be able to climb them. Climb all of them one after the other, with just a short break in between each problem. It's important that you focus on climbing them as perfectly as you can, adapting your style to match that of the problem. It's highly likely that precision footwork and balance will be key for the less-steep walls, while dynamics and direction of force will be important for the steeper climbs.



Marius Moe (2 metres tall) copying Sunniva Øvre Eide (1.69 metres).

COPYCAT

We all have our preferred climbing style, and in many instances it's either going to be the controlled, static style, or the more loose, dynamic style. A good exercise to challenge your familiar movement pattern is to climb a boulder problem in exactly the same way as your climbing partner. The bigger the difference between you and your partner, the more challenging the exercise will be. Ideally you will be at roughly the same level, but it's advantageous if your partner has differing strengths to you.

If you're a static climber, you'll learn a lot from copying a more dynamic climber. And in the same way, a tall climber can learn a lot by copying a short climber, as they'll then be challenged by compressed positions and high feet.



PROBLEM-SOLVING

When climbing, you're constantly having to solve problems: how to place your feet, how to grip the holds, whether you should move dynamically, whether you should twist in or front the wall. To become better at solving these problems, which are often split-second decisions, here's a good exercise. Pick five new boulder problems that have recently been set at your climbing wall and try to climb them all within 20 minutes. You should probably bring a stopwatch to keep track of time. You have an average of 4 minutes per problem and these minutes disappear quickly.

If there aren't any new problems at your wall, you can either visit a different wall or ask a friend to make up some problems for you.



TRAINING FOR TRAD & ALPINE

MARTINE LIMSTRAND is one of Norway's best trad and alpine climbers. She started out as a competition climber and has therefore logged countless hours of technique and strength training. We asked Martine to write a few words on how she feels these hours have helped her trad climbing.

Mastering climbing in alpine environments and placing your own gear is mostly about equipment and mental skills. Climbing past and falling on your own gear is difficult to practise without being outside, doing the work, placing the nuts and cams, and testing them. Still, there are many useful exercises you can do indoors to help you prepare. Generally speaking, improving your climbing skills will also make your mental game stronger, so by working on your footwork, finger strength and endurance, you can improve your trad climbing confidence.

Whether you wish to climb sustained cracks and corners, or you prefer varied multi-pitch climbing, having better endurance will always be an advantage. Alpine climbing often entails spending at least half an hour (or more) on a single pitch, and you're often gripping with the same hand for extended periods of time so you can place the right gear in the right spot. Even though you might not be able to practise placing gear indoors, it can be a good idea to do exercises indoors that improve your endurance and your ability to lock off.

Another element of alpine climbing is the long approaches. If you know you want to venture further into the outdoors, basic strength and endurance in your legs is important so that you have some strength and energy left over for the actual climbing. With proper technique you also use your legs and feet much more than your hands while climbing in alpine terrain. This you can practise indoors! Do exercises where you learn to trust bad footholds instead of pulling up with your arms. For example, try climbing slabs using just two or three fingers.

Remember: finger strength and slab climbing is more relevant than, for example, steep bouldering on big holds. Some climbing walls even have cracks that allow you to practise your jamming technique all through the winter months, which will come in handy!



PHOTO: OLA JAKOB DOVLAND

Martine enjoying herself at Fugløy, Norway.